Sensitivity-informed multi-objective adaptive weighting method for automatic alignment of freeform USTP systems

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Abstract: In catadioptric ultra-short-throw projection (USTP) systems, six-degree-of-freedom pose errors of the freeform mirror simultaneously affect geometric distortion and the modulation transfer function (MTF), causing significant coupling among multiple performance metrics during system alignment and leading to changes in the optimization objectives at different alignment stages. To address this problem, full-parameter six-degree-of-freedom scanning simulations were carried out to obtain the response laws of geometric distortion and MTF to pose perturbations. The results reveal that the dominant responses of these two metrics to pose errors undergo a stage-dependent shift as the error scale changes, and that stable mapping relationships exist between different distortion modes and specific subsets of degrees of freedom. On this basis, a physics-guided adaptive weight allocation and stage-wise optimization method is proposed. The above optical laws are explicitly incorporated into the weight evolution and optimization process, thereby establishing a physics-guided optimization framework for the automatic alignment of freeform USTP systems. Experimental results show that this method achieves stable convergence from coarse alignment to fine alignment. In 10 experiments with different initial conditions, the final mean MTF exceeded 70%, the alignment time for each run was about 10 min, and the dispersion of the results was significantly reduced. This work not only enables efficient automatic alignment of freeform USTP systems, but also provides a reference for modeling the coupling between pose errors and image-quality degradation and for designing alignment strategies in related optical systems.

1. Introduction

In precision optical systems, pose errors of optical elements are transferred to system-level imaging quality through aberrations, wavefront distortion, and geometric distortion. Therefore, the realization of a high-performance optical system depends not only on the optical design itself, but also heavily on the accuracy and stability of the subsequent alignment process [1]. This issue is particularly prominent in catadioptric ultra-short-throw projection (USTP) systems. Such systems usually adopt a compact folded optical path and a wide-field imaging structure [2,3]. The freeform mirror not only performs optical path folding [4], but also corrects large-scale aberrations and distortion [5]. Its six-degree-of-freedom (6-DOF) pose deviations simultaneously affect image geometric fidelity and the modulation transfer function (MTF), which causes the alignment process to exhibit significant strong coupling and nonlinear characteristics [6]. In engineering practice, the alignment of such systems still relies heavily on trial-and-error based on experience. This generally results in long cycle times, large result dispersion, and insufficient repeatability, which seriously restrict the efficient deployment and stable application of freeform USTP systems [7].

Optical system alignment usually relies on an iterative process among measurement feedback, performance evaluation, and pose correction. Its goal is to gradually approach the design state under limited adjustment degrees of freedom. Traditional optical alignment is usually centered on a single optimization objective. It drives element position correction by minimizing wavefront error, aberration terms, or a comprehensive image-quality function. In computer-aided alignment, researchers have further developed sensitivity-matrix-based methods. These methods estimate misalignment by establishing a linear mapping between measured quantities and compensation quantities [8,9], and combine the idea of balanced alignment to select highly sensitive compensation degrees of freedom for system-level error correction. In recent years, with the development of complex imaging systems, miniature camera modules, and space optical platforms, active alignment has gradually become an important direction of automatic alignment [10]. Compared with methods that rely on external high-precision metrology equipment, active alignment places more emphasis on using images, wavefronts, or system output performance as feedback, and directly performs error sensing and pose correction in the task domain [11]. Such methods have shown good engineering potential in various optical systems [12]. However, when the alignment target is further extended to off-axis, freeform, or large-initial-error scenarios, the local linear assumption is often no longer valid [13]. Problems such as ill-conditioned sensitivity matrices, error amplification, and unstable local convergence become much more severe [14].

For freeform USTP systems, the alignment result cannot be adequately characterized by a single metric [15]. Geometric distortion reflects the structural fidelity of the projected image, while MTF characterizes imaging sharpness and detail transfer capability. These two metrics correspond to different levels of image-quality requirements, and their dominant roles change with the error stage. Under large pose deviations, geometric distortion is usually the more prominent constraint. When the residual error becomes small, further improvement in system performance is more limited by fine imaging metrics such as MTF. Therefore, the automatic alignment of freeform USTP systems is essentially not a single-metric-driven error correction problem, but a multi-objective optimization problem with stage-dependent characteristics [16]. For multi-objective optimization, existing methods usually combine different image-quality metrics into a unified scalar objective through weight allocation [17]. Fixed-weight methods are simple to implement and have relatively strong interpretability, but they are difficult to adapt to changes in the dominant error sources and dominant performance metrics during alignment. To address this issue, researchers have proposed various adaptive weighting strategies, which dynamically adjust the importance of different objectives according to residuals, gradients, or iteration states [18]. Common forms of weight scheduling include linear scheduling, Sigmoid-type smooth transition, and update mechanisms based on the error variation rate [19]. Linear strategies are easy to implement, but they are sensitive to stage boundaries and are difficult to describe the true sensitivity variation in nonlinear systems [20]. Smooth functions such as the Sigmoid function can alleviate abrupt changes, but their transition position and slope usually depend on empirical settings and have limited physical interpretability [21]. Therefore, although these methods overcome the limitations of fixed weights, their weight evolution still mainly depends on predefined functions or empirical parameters, and has not fully reflected the true response laws of specific performance metrics to pose perturbations in a given optical system. For complex alignment problems that are strongly coupled, nonlinear, and involve stage-dependent migration of dominant objectives, this scheduling manner without a physical basis is often unable to balance early convergence efficiency and late-stage refinement capability [22]. In summary, for the automatic alignment of freeform USTP systems, existing studies still have the following shortcomings. First, there is a lack of systematic characterization of the coupling relationship among 6-DOF pose errors of the freeform mirror, geometric distortion, and MTF [23]. Second, there is a lack of clear understanding of how the dominance of different image-quality metrics changes at different error stages [24]. Third, there is still a lack of a specific alignment method that can further translate the above optical laws into degree-of-freedom decoupling organization, dynamic allocation of objective weights, and switching criteria for stage-wise optimization [25]. In other words, for this type of strongly coupled optical system, the key issue is not whether adaptive optimization should be used, but how to extract physical laws from the optical response of the system itself for alignment decision-making, and then construct an optimization strategy with physical interpretability on this basis.

To address the above problems, this paper first performs a systematic analysis of the responses of geometric distortion and MTF to 6-DOF pose perturbations of the freeform mirror through high-fidelity non-sequential ray-tracing simulation, and reveals the evolution laws of the dominance of different image-quality metrics under different error scales. Then, combined with the changes in typical distortion patterns induced by local perturbations, the dominant relationships between pose degrees of freedom and geometric degradation modes are extracted. On this basis, a physics-guided stage-wise adaptive alignment method is proposed, which unifies dynamic weight evolution, degree-of-freedom grouping, and closed-loop optimization execution within one framework, and is experimentally validated on a real freeform USTP platform.

The main contributions of this paper are as follows:

(1) Through high-fidelity non-sequential ray-tracing simulation, the response laws of geometric distortion and MTF to 6-DOF pose errors of the freeform mirror in a freeform USTP system are systematically characterized, and the stage-dependent shift in the dominance of different image-quality metrics with error scale is revealed.

(2) Based on the analysis of image degradation patterns induced by local perturbations, the dominant mapping relationships between typical geometric distortion modes and subsets of pose degrees of freedom are established, which provides a physical basis for degree-of-freedom grouping and local optimization.

(3) A physics-guided stage-wise adaptive alignment method is proposed. It explicitly incorporates optical laws into the weight evolution and optimization process, achieves stable convergence from coarse alignment to fine alignment, and is validated on a real platform in terms of efficiency, stability, and repeatability.

The remainder of this paper is organized as follows. Section 2 presents the simulation analysis of the pose-image-quality mapping relationship and the discovery of physical laws. Section 3 proposes a hierarchical dynamic weighting optimization method driven by simulation experience. Section 4 verifies the effectiveness of the proposed method through comparative experiments. Section 5 summarizes the paper and discusses future research directions.

2. Analysis of the simulated pose–image-quality relationship

The effect of freeform mirror pose errors on the imaging performance of the USTP system shows clear multidimensional coupling. Therefore, the magnitude of image-quality degradation alone is not sufficient to guide the subsequent alignment strategy. In this section, a high-fidelity non-sequential optomechanical model, consistent with the physical prototype, is used to systematically investigate the responses of geometric distortion and MTF to six-degree-of-freedom perturbations of the freeform mirror. The analysis focuses on how the dominant image-quality factors vary with error magnitude, and on the dominant correspondence between typical geometric distortion patterns and specific pose degrees of freedom. These results provide the physical basis for the subsequent stage-wise optimization, metric weight scheduling, and degree-of-freedom grouping.

2.1 Construction of a ray tracing simulation model in non-sequential mode

In engineering implementation, USTP systems usually adopt an assembly process that combines modular pre-assembly with subsequent compensatory alignment. For the prototype studied in this work, the refractive lens group and some fixed reflective components are usually positioned and initially calibrated during assembly. Under mass-production conditions, their residual errors mainly appear as relatively stable initial system offsets. In contrast, the freeform mirror, which is responsible for folded optical path formation and wide-field image-quality correction, is connected to the mechanical frame through an adjustable mechanism. Its pose is more sensitive to system geometric distortion and MTF, and it is also the most important adjustable optical element in the subsequent alignment process [26,27]. Based on prior tolerance analysis and optomechanical structure design, the automatic alignment problem in this paper is modeled as a pose optimization problem in which the freeform mirror serves as the primary compensating element. Specifically, this work focuses on the influence of six-degree-of-freedom pose errors of this mirror on system imaging quality and the corresponding compensation mechanism, while the residual errors of the other components are uniformly incorporated into the initial system state. On the one hand, this modeling method is consistent with the actual alignment procedure of the prototype studied here. On the other hand, it also helps emphasize the dominant role of pose errors of the key element in the overall imaging quality.

A complete simulation model of the USTP optical system is established in the non-sequential mode of Ansys Zemax OpticStudio. As illustrated in Fig. 1, the model is constructed in strict accordance with the actual optical design and incorporates the light source, the front and rear refractive groups, the freeform mirror, and the projection image plane. A preliminary comparison between the simulation results and experimental data obtained from the prototype indicates consistent dominant distortion patterns and similar relative variation trends, thereby confirming the reliability of the model and providing a credible basis for subsequent analyses of image-quality behavior under pose perturbations. Within this model, all optical components are assumed to be ideally aligned except for the freeform mirror, which is treated as the only adjustable compensator. Under this setting, to systematically investigate how mirror pose errors influence imaging quality, controllable pose perturbations are introduced into the simulation model. This is achieved by inserting coordinate breaks before and after the mirror, which allows six-degree-of-freedom rigid-body motions to be applied precisely in its local coordinate system, including translations along X, Y, and Z and rotations about A, B, and C, while preserving the global optical layout. This approach effectively avoids errors arising from coordinate transformations and ensures both the accuracy and repeatability of the simulation results. The scanning ranges of the simulation parameters are summarized in Table 1 and are chosen to cover and slightly exceed the practical alignment tolerances, thereby ensuring the completeness and engineering relevance of the study.

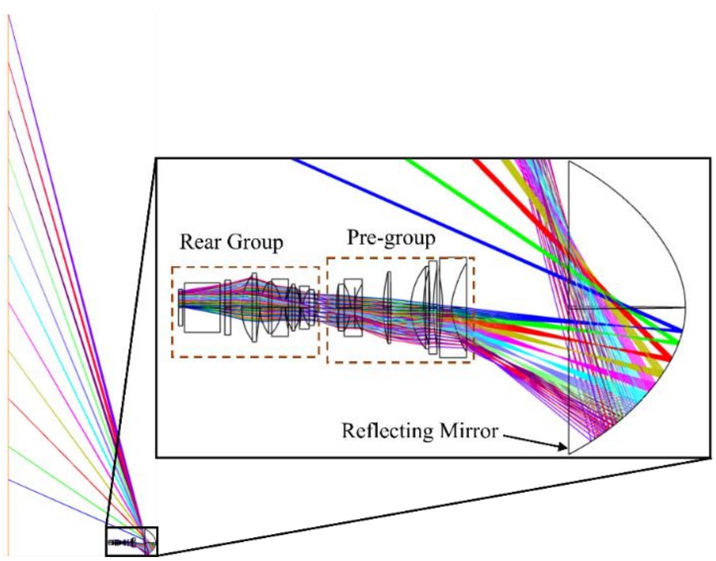

Fig. 1. Designed structure of the USTP optical system.

Table 1. Simulation parameters for 6-DOF

Degree of freedom	Unit	Scan range	Step
X	mm	[-10, 10]	0.05
Y	mm	[-10, 10]	0.05
Z	mm	[-10, 10]	0.05
A	deg	[-10, 10]	0.05
B	deg	[-10, 10]	0.05
C	deg	[-10, 10]	0.05

2.2 Image quality evaluation metrics

For USTP systems, image quality is commonly evaluated in terms of MTF and distortion. In this work, the sharpness of the projected image is quantified by the MTF value at a target spatial frequency of 93 lp/mm. According to industrial practice, geometric distortion can be further decomposed into three key components, namely fan distortion (D_f), tilt distortion (D_t) and keystone distortion (D_k). Among them, D_f can be regarded as a component-wise description of conventional radial distortion in terms of local boundary morphology, and it reflects the local fan-like expansion of barrel distortion, pincushion distortion, and their higher-order waveform distortions. D_t can be regarded as a tilted variant of tangential (decentering) distortion under asymmetric projection conditions, whereas D_k corresponds to the keystone distortion commonly observed in projection systems. As illustrated in Fig. 2, these three components characterize different types of edge deformation and have a significant impact on the accuracy of the imaging system. Specifically, D_f measures the deviation of the mid-point of an image edge from the straight line connecting its two endpoints, D_t denotes the deviation of the two endpoints of an image edge from the vertical direction in the reference rectangular coordinate system, and D_k describes the length difference between the upper and lower edges of the image, which reflects the degree of keystone deformation. The corresponding formulas are given as follows:

$$\begin{cases} D_f = E_y - E_{0y} \\ D_t = A_y - B_y \\ D_k = |AB| - |CD| \end{cases} \tag{1}$$

where E_y denotes the actual coordinate of the midpoint of the image edge and E_{0y} is the ideal coordinate. A_y and B_y represent the vertical coordinate differences at the two endpoints of the image edge. $|AB|$ and $|CD|$ denote the lengths of the upper and lower edges of the image, respectively.

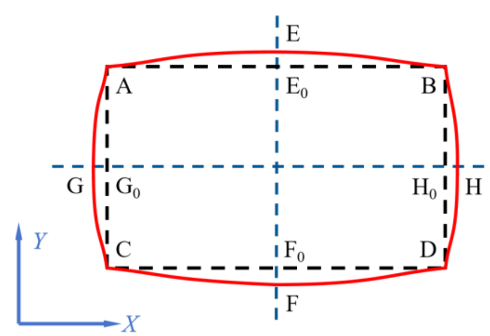

Fig. 2. Deformation description.

To obtain a unified evaluation domain for both simulation and experiment, a projection test pattern was designed as shown in Fig. 3. The pattern consists of an MTF test array, a distortion calibration frame, and a crosshair formed by fiducial blocks. In this pattern, the corner points of the outer frame are used to measure and compute the three types of distortion. The MTF test array is composed of cells with different line-pair patterns, and the overall layout follows the structure of the 1951 USAF resolution chart, providing horizontal and vertical line pairs at multiple spatial frequencies. In this work, the group corresponding to 93 lp/mm is selected, and the MTF at this frequency is obtained from the contrast of the stripes in the simulated or captured images.

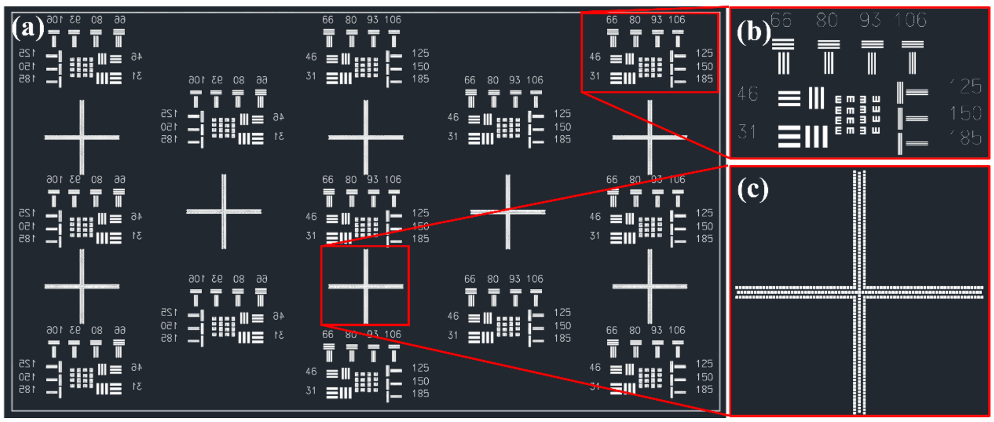

Fig. 3. Projection test chart for MTF and distortion evaluation. (a) Overall test pattern, consisting of an MTF measurement array and crosshair alignment markers, used to uniformly evaluate the spatial resolution and geometric distortion of the imaging system over the full field of view; (b) enlarged view of the MTF test element, designed based on the USAF 1951 resolution target, containing horizontal and vertical bar patterns with different spatial frequencies; (c) enlarged view of the crosshair alignment marker.

2.3 Global sensitivity analysis of image quality with respect to pose perturbations

By performing a full-range scan over all six degrees of freedom, we obtained the variation curves of MTF and geometric distortion with respect to pose perturbations, where the latter is quantified by a composite distortion metric. Here, the composite distortion metric D_c is defined as $D_c = \sum_{i=1}^{n} \omega_i D_i$ where D_i denotes the normalized value of the i-th distortion component and ω_i is its corresponding weight. In the sensitivity analysis of the 6-DOF, denoted by $q \in \{X, Y, Z, A, B, C\}$, each degree of freedom is scanned across its full range as specified in Table 1. At each sampling point, the corresponding composite geometric distortion $D(q)$ and the MTF evaluated at a spatial frequency of 93 lp/mm, $MTF(q)$, are calculated. To facilitate comparison of both magnitude and slope among different degrees of freedom, each distortion curve is normalized by the maximum distortion value observed within the scan range of the corresponding degree of freedom, i.e.,

$$\hat{D}(q) = \frac{|D(q)|}{\max_q |D(q)|} \in [0,1] \tag{2}$$

where $\hat{D}(q) = 1$ denotes the maximum distortion over the full scan range of the corresponding degree of freedom. MTF is evaluated using the contrast method, by analyzing the gray-level variation in the image.

$$MTF(f) = \frac{\mathrm{C}(f)}{\mathrm{C}_0} \tag{3}$$

where $C(f)$ is the contrast at spatial frequency f and C_0 is the contrast at low frequency. The contrast C is computed from the maximum and minimum gray levels, I_{max} and I_{min}, as

$$\mathrm{C} = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} \tag{4}$$

To quantitatively evaluate the influence of freeform mirror pose errors on image quality, a global sensitivity analysis map is constructed, as illustrated in Fig. 4. In this map, the horizontal axis represents the pose perturbation of each degree of freedom, with millimeters used for the translational DOFs (*X, Y, Z*) and degrees for the rotational DOFs (*A, B, C*). The vertical axis simultaneously presents two key image-quality metrics, namely the normalized composite geometric distortion and the normalized MTF. By scanning all 6-DOF across their full ranges, a complete mapping between pose errors and image-quality degradation is obtained.

Further analysis of these curves shows that the sensitivities of geometric distortion and MTF to pose errors are not constant over the full perturbation range. In this work, this phenomenon is referred to as stage-dependent sensitivity, meaning that the response strengths of image-quality metrics to 6-DOF pose perturbations vary with the error magnitude, and that the relative importance of different metrics may change across different alignment stages. Taking the tilt DOF A in Fig. 4(a_2) as an example, within the large-deviation region of $|A| \in [2°, 10°]$, corresponding to the coarse alignment stage, the normalized distortion increases sharply from near 0 to 1, whereas the normalized MTF decreases only gradually from approximately 0.7 to 0.5. This behavior indicates that, during coarse alignment, pose perturbations have a much stronger influence on geometric distortion than on MTF, with their rates of change differing on the order of magnitude. At this stage, image-quality degradation is therefore dominated by geometric deformation of the projected pattern, and the optimization strategy should prioritize suppressing distortion to recover the basic geometric structure of the image.

As the error range is further reduced to $|A| \leq 1°$, corresponding to the fine alignment stage, the distortion curve gradually saturates and its variation rate decreases significantly, approaching the design tolerance and leaving limited room for further improvement. In contrast, the MTF curve remains highly sensitive within this region, exhibiting noticeable changes even under small pose adjustments. This observation implies that, in the fine alignment stage, the dominant factor limiting further image-quality improvement shifts from geometric deformation to image sharpness. Accordingly, the optimization objective should transition toward fine tuning of the MTF in order to fully exploit the ultimate performance potential of the system.

The above global sensitivity analysis shows that the image-quality degradation caused by pose perturbations is not always dominated by a single metric, but exhibits clear stage-dependent characteristics. Over a large error range, geometric distortion increases rapidly with pose deviation, whereas the variation in MTF is relatively moderate. In this stage, image degradation is mainly manifested as geometric structural distortion. As the pose error is further reduced, the response of geometric distortion gradually approaches saturation, while MTF remains highly sensitive to small perturbations. Under this condition, further improvement in system performance is more strongly limited by imaging sharpness. These results indicate that geometric distortion and MTF play different dominant roles in different error ranges, which provides an analytical basis for the subsequent development of stage-wise evaluation and optimization strategies.

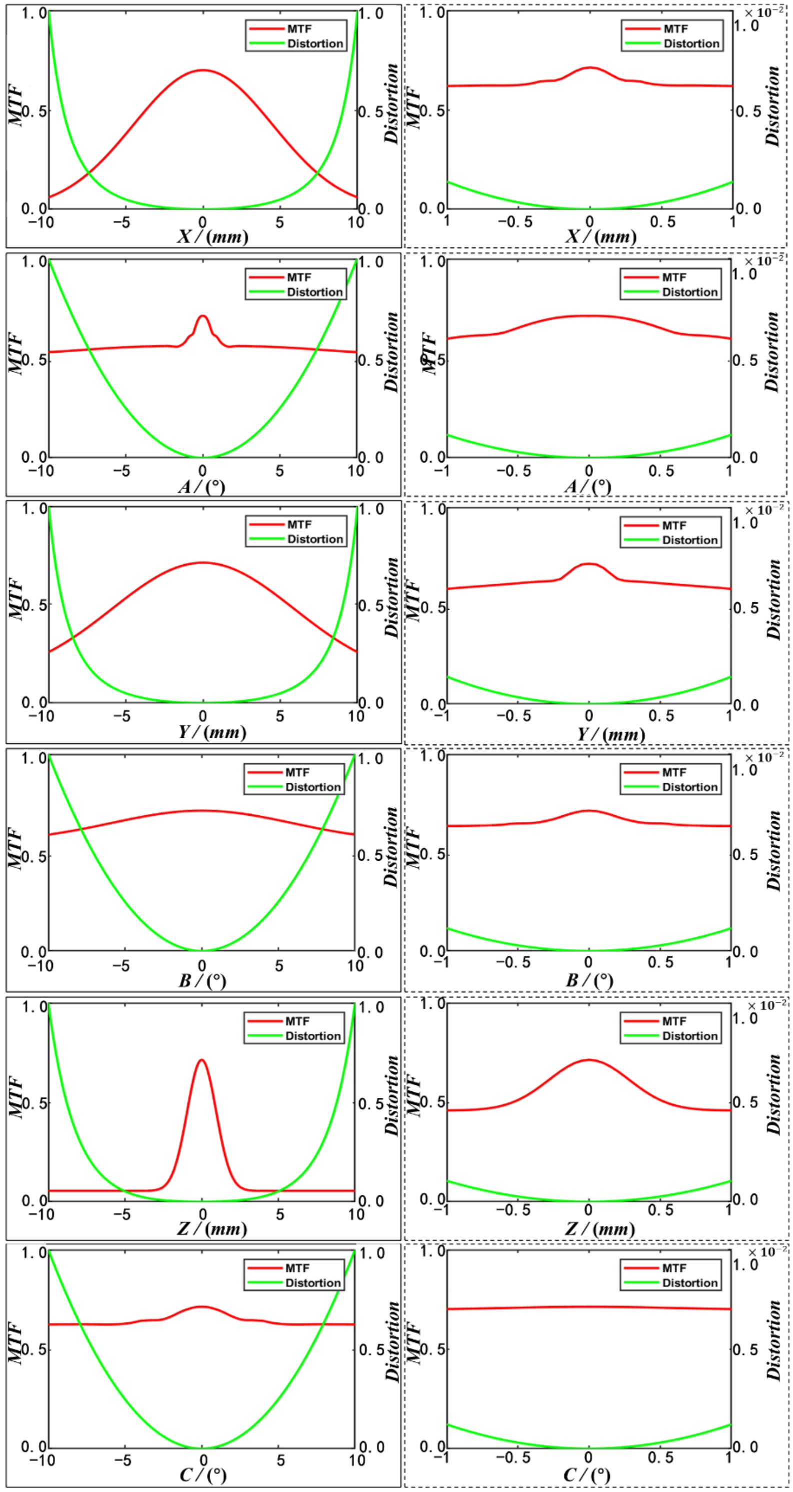

Fig. 4. Global sensitivity of normalized distortion and MTF to 6-DOF mirror pose perturbations. (a_1-a_6) shows sensitivity curves of normalized distortion and MTF from full range scanning. (b_1-b_6) shows sensitivity curves of normalized distortion and MTF from fine range scanning.

2.4 Symmetry-based decoupling of geometric distortion and DOF correlation analysis

To further reveal the influence of different degrees of freedom (DOFs) on geometric distortion modes, image formation under local pose perturbations of the freeform mirror is examined using a standard projection test pattern within the non-sequential simulation framework. With all other parameters fixed, each DOF is independently perturbed by small positive and negative increments (±1 mm for translations and ±1° for rotations), and the corresponding projected images are analyzed. The results shown in Fig. 5 indicate that perturbations in the DOF subset $\{X, A, C\}$ mainly induce odd-symmetric distortion, manifested as a global tilt of the projected image, while fan-shaped and keystone distortions remain relatively weak. In contrast, perturbations in the subset $\{Y, B, Z\}$ predominantly excite even-symmetric distortion, characterized by keystone deformation accompanied by fan-shaped bulging or indentation near the image boundaries. These behaviors are consistent with the distinct physical roles of the two subsets in controlling ray incidence direction versus mirror height, tilt, and image-side focal position.

The above local perturbation results indicate that, within the alignment range studied here, stable dominant relationships exist between different geometric distortion modes and several subsets of pose degrees of freedom, as summarized in Table 2. Specifically, odd-symmetric tilt-like distortion is mainly influenced by the $\{X, A, C\}$ subspace, whereas even-symmetric fan-shaped and keystone distortion are more strongly modulated by the $\{Y, B, Z\}$ subspace. Based on this correlation, the six-degree-of-freedom pose optimization can be further organized into two search subspaces with different emphases. This improves the directional relevance of the search and provides an analytical basis for the subsequent local weight allocation and grouped optimization.

Table 2. Dominant relationships between degree-of-freedom subsets and distortion modes.

DOF subset	Dominant distortion type(s)	Correlation strength
{X, A, C}	Tilt distortion	Strong
{Y, B, Z}	Fan distortion, keystone distortion	Strong

The above coupled perturbation results further verify the grouping relationships obtained from the single-degree-of-freedom analysis. Fig. 6 supplements several typical coupled perturbation cases. (a_1) and (b_1) correspond to coupled perturbations in $\{X, A, C\}$, and the resulting images are both dominated by overall tilt-like distortion. (a_2) and (b_2) correspond to coupled perturbations in $\{Y, B, Z\}$, which mainly lead to keystone and fan-shaped distortion, with the distortion direction reversing as the sign of B changes. (a_3) and (b_3) further show the superposition of these two types of distortion features under cross-degree-of-freedom combinations. These results indicate that, under coupled perturbation conditions, the dominant distortion trends of each degree-of-freedom subset remain basically consistent with those identified in the single-degree-of-freedom analysis.

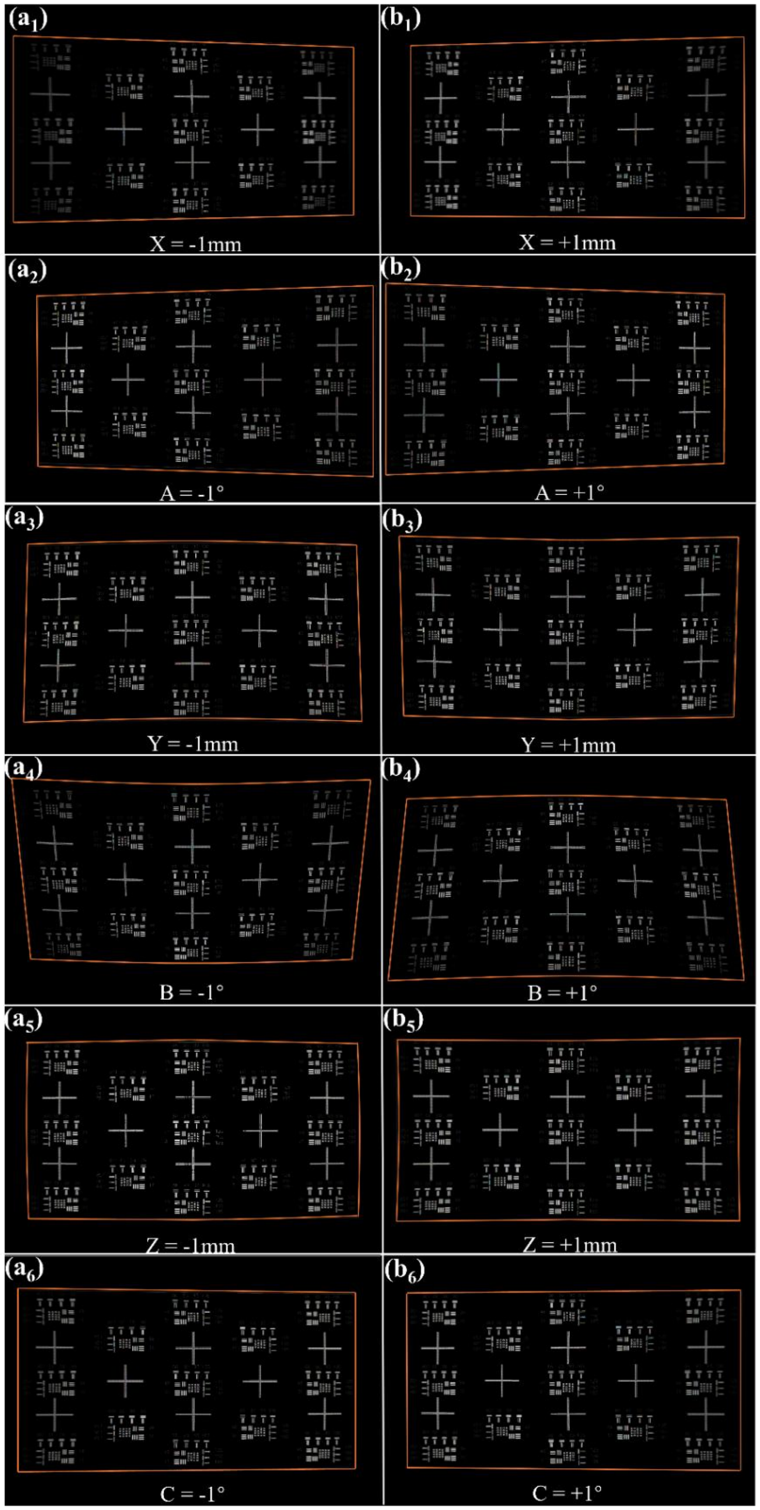

Fig. 5. Local distortion patterns induced by {X, A, C} and {Y, B, Z} pose perturbations. (a_1) X = −1 mm, (b_1) X = +1 mm; (a_2) A = −1°, (b_2) A = +1°; (a_3) Y = −1 mm, (b_3) Y = +1 mm; (a_4) B = −1°, (b_4) B = +1°; (a_5) Z = −1 mm, (b_5) Z = +1 mm; (a_6) C = −1°, (b_6) C = +1°.

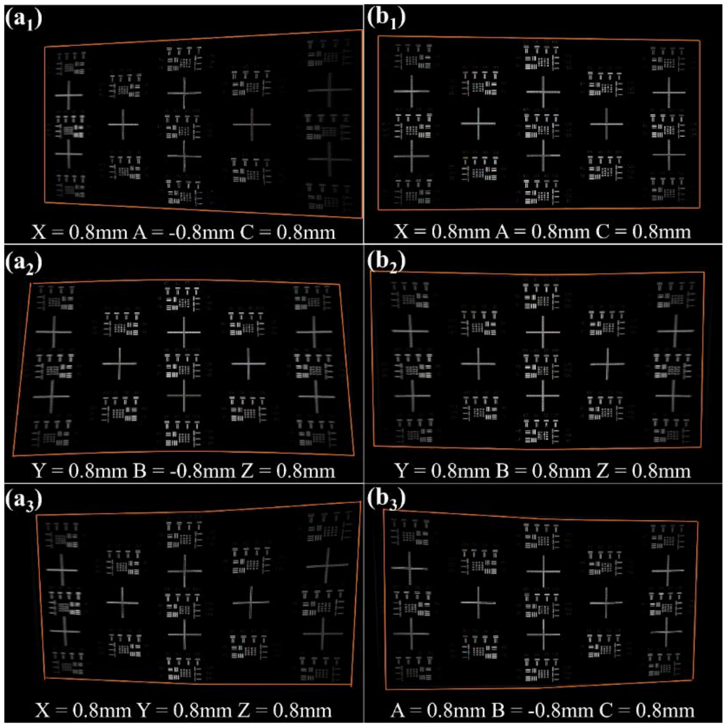

Fig. 6. Coupled distortion patterns induced by representative multi-DOF perturbations of {X, A, C}, {Y, B, Z}, and cross-group pose combinations. (a1) X = 0.8 mm, A = −0.8°, C = 0.8°; (b1) X = 0.8 mm, A = 0.8°, C = 0.8°; (a2) Y = 0.8 mm, B = −0.8°, Z = 0.8 mm; (b2) Y = 0.8 mm, B = 0.8°, Z = 0.8 mm; (a3) X = 0.8 mm, Y = 0.8 mm, Z = 0.8 mm; (b3) A = 0.8°, B = −0.8°, C = 0.8°.

3. A Physically-Guided Hierarchical Dynamic Weighting Strategy

3.1 Strategic Framework Overview

As shown in Fig. 7, the proposed physics-guided adaptive weighting scheme consists of three parts: simulation analysis, optimization design, and closed-loop execution. Based on the pose-image-quality relationship obtained in Section 2, the method introduces both a global stage-wise weight scheduling strategy and a local grouped optimization mechanism into the optimization process. The stage transition is determined according to the dominant changes of geometric distortion and MTF over different error ranges. The normalization factors are set by combining optical sensitivity and alignment tolerance. The local optimization is organized according to the relationships between degree-of-freedom subsets and typical distortion modes.

In the system analysis and modeling stage, the projection optical system and its image-quality evaluation model are first established. On this basis, a simulation model is constructed to characterize the response relationship between spatial pose errors and image-quality metrics. In the optimization function design stage, the relative weights of geometric distortion and MTF are dynamically adjusted through a global weighting strategy. At the same time, a local grouping mechanism is introduced to enhance the optimization specificity along different degree-of-freedom directions, thereby forming a unified scalar objective function $J(\theta, \omega)$. In the closed-loop execution stage, the system uses an industrial camera to capture projection images and complete metric evaluation. The multi-stage dynamic weighting optimizer then generates control commands to drive the six-degree-of-freedom motion platform for pose adjustment. Through iterative updating with real-time feedback, the process continues until the convergence condition is satisfied and automatic alignment is completed.

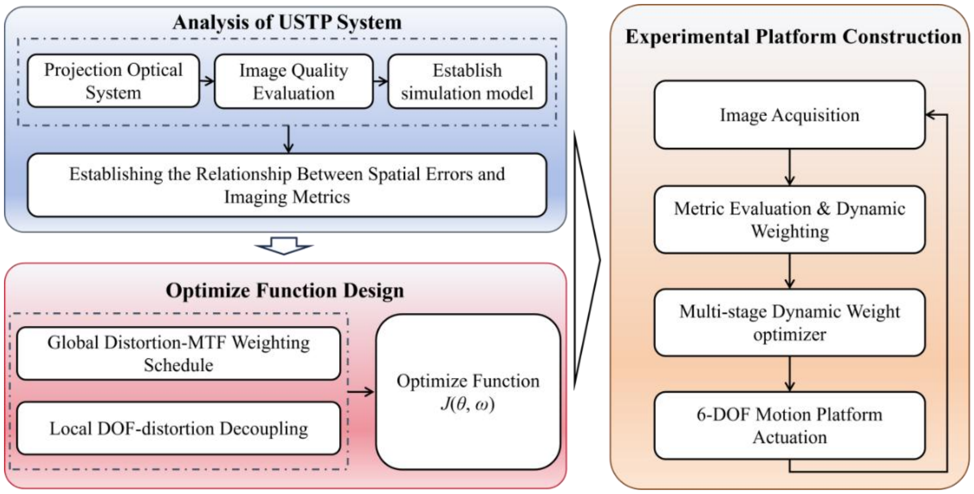

Fig. 7. Physics-Guided Simulation-Optimization-Execution Loop Flowchart.

3.2 Construction of a Scalar Evaluation Function

To reduce distortion in the projected image as much as possible during assisted alignment while at the same time maintaining image sharpness, both geometric distortion and image quality are jointly evaluated over the entire field. Let the mirror pose be denoted by

$$\theta = [x, y, z, a, b, c]^T \in \Omega \tag{5}$$

where (x, y, z) are translations, (a, b, c) are small rotations about the (X, Y, Z) axes, and Ω is the feasible domain determined by the mechanical structure and alignment tolerances. Using the structured projection pattern as observation, three categories of geometric metrics - D_f, D_t, D_k, and MTF - are defined. To account for spatial variation, all metrics are statistically evaluated over a set S of regions of interest.

To eliminate the differences in dimension and scale among different metrics, the normalization factor was set to $\tau_* = 1.35$. This value is not an arbitrary numerical scaling factor. Instead, it is based on the dimensionless characteristic scale defined by the optical sensitivity features identified in Section 2.3 and the prescribed alignment tolerance. Its physical meaning is that, when the normalized residual exceeds about 1.35 characteristic units, the Huber robust kernel $\rho(\cdot)$ suppresses local outliers, thereby

$$\begin{cases} \hat{d}_{D_t}^{(s)} = \rho(\dfrac{D_t^{(S)}}{\tau_{D_t}}) \\ \hat{d}_{D_f}^{(s)} = \rho(\dfrac{D_f^{(S)}}{\tau_{D_f}}) \\ \hat{d}_{D_k}^{(s)} = \rho(\dfrac{D_k^{(S)}}{\tau_{D_k}}) \\ \hat{m}^{(s)} = \rho(\max(0, 1 - \dfrac{MTF^{(S)}}{\tau_{MTF}})) \end{cases} , s \in S \tag{6}$$

For quantitative evaluation, the objective function is defined as:

$$J(\theta; \omega) = \sum_{s \in S} (\omega_{D_t} \hat{d}_{D_t}^{(S)} + \omega_{D_f} \hat{d}_{D_f}^{(S)} + \omega_{D_k} \hat{d}_{D_k}^{(S)} + \omega_{MTF} \hat{m}_{MTF}^{(S)}) \tag{7}$$

where $\Sigma\omega_* = 1$ and $\omega_* \geq 0$. We then seek to minimize J over the feasible domain Ω.

3.3 Physically Guided Weight Allocation Criteria

Based on the sensitivity analysis from pose to image quality in Section 2.3 and practical alignment experience, the focus on image-quality metrics varies significantly across different stages within the adjustable working domain. In the coarse alignment stage, the primary objective is to rapidly suppress pronounced geometric distortion in order to restore the basic structure integrity of the image. In the fine-tuning and verification stages, the emphasis shifts to further improving and balancing image sharpness while maintaining the established geometric shape. To avoid frequent manual adjustment of evaluation weights during optimization, we adopt a sensitivity-aware continuous adaptive weighting strategy initialized with a single set of weights. During optimization, the weights are smoothly updated according to the residual variations of individual metrics, allowing them to gradually migrate as the optimization processes and the sensitivities evolve. This enables a natural transition from a geometric-distortion-dominated regime, to a balanced consideration of distortion and sharpness, and finally to increased emphasis on sharpness consistency.

To translate this macroscopic strategy into a reproducible scalar formulation, we construct a two-level weighting structure with a global layer and a local layer. At the global layer, the overall weights assigned to the geometric distortion group and the MTF group are balanced. At the local layer, the distortion weight is further decomposed into three specific indices, D_f, D_t and D_k, thereby enabling fine-grained weight management from macroscopic control to microscopic refinement.

3.3.1 Global adaptive weighting design

At the global level, we introduce two non-negative weights $\omega_{dist}(k)$ and $\omega_{MTF}(k)$ such that

$$\begin{cases} \omega_{dist}(k) = \omega_{D_t}(k) + \omega_{D_f}(k) + \omega_{D_k}(k) \\ \omega_{dist}(k) + \omega_{MTF}(k) = 1 \end{cases} \tag{8}$$

where $\omega_{dist}(k)$ represents the proportion of geometric distortion in the overall evaluation at iteration and $\omega_{MTF}(k)$ represents the relative weight assigned to the sharpness metric.

During the iterative process, larger variations in a given distortion metric indicate higher error sensitivity along the corresponding degree-of-freedom direction. Accordingly, the associated weight is increased to stabilize the optimization process. To characterize the variation of the distortion metric, its rate of change is defined as

$$r_i(k) = \frac{\left| d_i^{(s)}(k) - d_i^{(s)}(k-1) \right|}{d_i^{(s)}(k-1) + \varepsilon} \tag{9}$$

where k is the iteration index and ε is a small positive constant introduced to avoid division by zero. The rate $r_i(k)$ characterizes the instantaneous sensitivity of the system to the i metric. The rates are then normalized as

$$\bar{r}_i(k) = \frac{r_i(k)}{\sum r_j(k)} \tag{10}$$

To ensure that the weight adjustment both reflects changes in sensitivity and remains smooth and continuous, an exponential-type adaptive update rule is employed,

$$\begin{cases} \omega_i(k) = \dfrac{e^{\alpha \bar{r}_i(k)}}{\sum e^{\alpha \bar{r}_i(k)}} \\ \omega_i(k) = (1-\beta)\omega_i(k-1) + \beta \omega_i(k) \end{cases} \tag{11}$$

where $\alpha > 0$ is a tuning coefficient that controls the amplification of sensitivity, and $\beta \in (0,1]$ is a smoothing coefficient used to suppress high-frequency oscillations and ensure trajectory continuity. In the experiments, we set $\alpha = 2.0$ and $\beta = 0.85$. An exponential update rate was adopted so that the exponential mapping can enhance the response to highly sensitive metrics while ensuring that the weights remain non-negative and can be normalized. As a result, the weight allocation has a stronger ability to focus on dominant errors. In addition, the continuous and smooth functional form is also beneficial for reducing weight oscillation and maintaining the stability of the optimization trajectory. When the rate of change of a metric is large at a given iteration, the corresponding update factor $\omega_i(k)$ becomes significantly larger, thereby increasing its weight and achieving adaptive enhancement of that metric. For metrics that vary more slowly, their weights are relatively reduced. This mechanism naturally produces an adaptive shift of optimization emphasis, so that at different stages the algorithm automatically allocates greater attention to the currently dominant error source.

3.3.2 Local adaptive weighting design

After the global weights are determined, the distortion weight is further distributed to D_f, D_t, and D_k, according to the correlation between distortion modes and degree-of-freedom subsets obtained in Section 2.4, thereby constructing the local distortion weight vector.

$$\omega_{loc}(k)=[\omega_{D_t}(k),\ \omega_{D_f}(k),\ \omega_{D_k}(k)]^T \tag{12}$$

where k denotes the iteration index. For consistency with the global layer, the local weights are required to satisfy:

$$\omega_{D_t}(k)+\omega_{D_f}(k)+\omega_{D_k}(k)=\omega_{dist}(k) \tag{13}$$

According to the sensitivity distribution, the DOF subset $\{X, A, C\}$ mainly induces D_t, whereas the subset $\{Y, B, Z\}$ primarily excites D_f and D_k. Based on this observation, two normalized local weight templates are predefined for these two DOF groups,

$$\begin{cases} b^{(XAC)}=[b_{D_t}^{(XAC)},b_{D_f}^{(XAC)},b_{D_k}^{(XAC)}]^T \\ b^{(YBZ)}=[b_{D_t}^{(YBZ)},b_{D_f}^{(YBZ)},b_{D_k}^{(YBZ)}]^T \end{cases} \tag{14}$$

where $\sum_i b_i^{(XAC)}=\sum_i b_i^{(YBZ)}=1$, and specify the desired relative proportions of D_f, D_t, and D_k for the odd DOF subset $\{X, A, C\}$ and the even DOF subset $\{Y, B, Z\}$, respectively. In a typical setting, the template $b^{(XAC)}$ assigns a larger share of the distortion weight to D_t, while $b^{(YBZ)}$ slightly reinforces D_f and D_k. In physical terms, this realizes an alignment rhythm in which the subset $\{X, A, C\}$ is first used to rapidly suppress the dominant odd symmetric distortion, followed by the subset $\{Y, B, Z\}$ to drive the residual even symmetric components to convergence.

To enable dynamic allocation of local weights, a group indicator $\delta_{XAC}(k)$ is introduced to identify which DOF subset is being predominantly optimized at iteration k. When the k-th iteration mainly targets odd symmetric distortion, $\delta_{XAC}(k) = 1$; otherwise, $\delta_{XAC}(k) = 0$. The local target weight vector at iteration k is then defined:

$$\omega_i(k)=\omega_{dist}(k)[\delta_{XAC}(k)b_i^{(XAC)}(k)+(1-\delta_{XAC}(k))b_i^{(YBZ)}(k)],\ i\in\{D_t,D_f,D_k\} \tag{15}$$

This mechanism ensures that the sum of the three distortion weights remains conserved at every iteration, and that the allocation of distortion weight automatically switches between the $\{X, A, C\}$ and $\{Y, B, Z\}$ subsets. When optimization is carried out in the $\{X, A, C\}$ subspace, the weights concentrate on the distortion component it predominantly controls; when the process switches to the $\{Y, B, Z\}$ subspace, the weights shift to the distortion components dominated by this group.

4. Experimental Validation and Discussion

4.1 Experimental Setup

To achieve fast and high-precision alignment of the USTP system, a closed-loop alignment setup based on a 6-DOF motion stage is constructed, as shown in Fig. 8. The system consists of a USTP module, a high-precision industrial camera and a 6-DOF motion stage. The stage provides pitch, yaw and axial translation, among other motions, and compensates the projected image by precisely adjusting the pose of the freeform mirror. The optical engine is installed in a darkroom environment to suppress ambient light, and geometric and photometric calibration are carried out to remove lens distortion of the camera and to establish a pixel-level spatial mapping between the camera and the projector. During alignment, the mirror serves as the only dynamically controlled component, and its pose parameters are iteratively updated using a hybrid optimization strategy that combines Levenberg–Marquardt for global convergence with simulated annealing for local perturbations. The final alignment target is that all ten distortion components (four D_t terms, four D_f terms and two D_k terms) are reduced below 30 pixels, while the MTF at 93 lp/mm exceeds 70%.

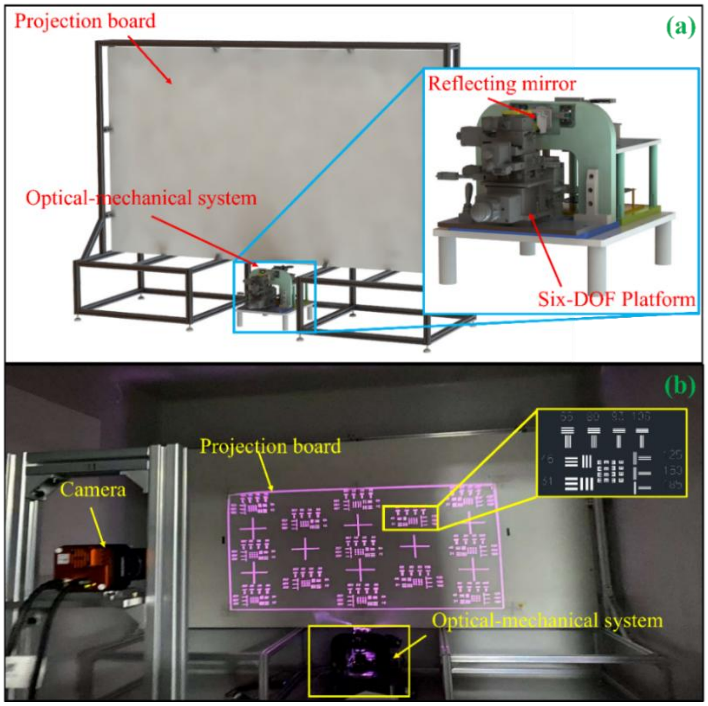

Fig. 8. USTP mirror-assisted alignment system. (a) Schematic diagram of system architecture. (b) Experimental platform setup

To cover the full projected field of about 100 inches (16:9, approximately 2.21 m × 1.25 m), the camera is placed at a distance of about 2 m from the screen, and a Hikrobot MV-CH650-90TM/TC industrial camera is used. Its high resolution and 3.2 μm pixel pitch provide sufficient spatial sampling at this distance and image size (about 10 pixels per line pair at 93 lp/mm). The global shutter eliminates geometric artifacts caused by motion and timing, while the dynamic range together with programmable exposure and gain ensures stable imaging under different brightness conditions. The high-speed industrial interface also satisfies the bandwidth requirements of the closed-loop evaluation. The detailed specifications of the camera are listed in Table 3.

Table 3. Key camera parameters

Parameter	Specification
Sensor type	CMOS, global shutter (Gpixel GMAX3265)
Resolution	9344 × 7000
Pixel size	3.2 μm × 3.2 μm
Max frame rate	≈17.2 fps (mode-/interface-dependent)
Exposure time	18 μs – 10 s, programmable (auto exposure supported)
Analog/Digital gain	0 – 24 dB adjustable

To ensure stable platform execution, constraints were imposed on both the single-step pose increment and the cumulative search range during optimization. In the coarse search stage, the upper limits of the translational and rotational step sizes were 0.50 mm and 0.50°, respectively. In the fine alignment stage, they were 0.05 mm and 0.05°, respectively. The overall search was always restricted within the range of $|x|$, $|y|$, $|z| \leq 10$mm and $|a|$, $|b|$, $|c| \leq 10°$. In the experiments, the captured images were first corrected by calibration, and then the outer-frame corner points, edge midpoints, and the 93 lp/mm stripe region were extracted. A detailed illustration of the image-processing workflow used for ROI extraction, coarse/fine localization, and MTF recognition is provided in the Supplementary Material (Fig. S1). The three types of distortion and the MTF were then calculated according to Eqs. (1)–(4). These quantities were further normalized by Eq. (6) and finally combined through Eq. (7) to construct the comprehensive evaluation value for closed-loop optimization.

4.2 Ablation experiments: validation of the internal mechanism of the hierarchical weighting scheme

To further investigate the internal mechanism of the hierarchical weighting strategy, a series of ablation experiments was designed in this study. First, to evaluate the overall advantage of the proposed physics-guided adaptive weight allocation scheme, this subsection compared our method with two representative adaptive weighting methods and three fixed-weight schemes under the same initial pose, the same evaluation threshold, and the same optimization conditions. In the present study, a simulated annealing strategy was adopted during the coarse-search stage, with a geometric cooling schedule, where the temperature was updated as $T_n=0.95^n T_0$, and inferior solutions were accepted with a probability following the Metropolis criterion, $p=exp(-\Delta U/T)$. To ensure reproducibility and fairness of comparison, all experiments in this subsection used the same optimization settings. Early stopping was triggered when the number of iterations reached 1000 or when the composite evaluation value fell below 50. In addition to the three fixed-weight schemes, this subsection also included two representative dynamic weighting methods as comparisons, namely the linear adaptive weighting scheme and the Sigmoid adaptive weighting scheme. All schemes were initialized with the same weight vector [0.30, 0.30, 0.30, 0.10], where the four components correspond to D_f, D_t, and D_k, and MTF, respectively.

For the linear adaptive weighting method, the weight update is defined as

$$\begin{cases} \overset{lin}{\omega_i}(k) = 1 + \alpha_{lin} r_i(k) \\ \omega_i^{lin}(k) = (1-\beta_{lin})\omega_i^{lin}(k-1) + \beta_{lin}\overset{lin}{\omega_i}(k) \end{cases} \tag{16}$$

where $\alpha_{lin}=1.0, \beta_{lin}=0.7$. These values were chosen because the linear method belongs to constant-slope regulation. If the gain is too large, overshoot can easily occur near the stage boundary. Therefore, a moderate gain and a weaker historical inertia term were used here to reflect its linear transition characteristics.

For the Sigmoid adaptive weighting method, its update form is defined as

$$\begin{cases} \sigma(x) = \dfrac{1}{1+e^{-x}} \\ \overset{sig}{\omega_i}(k) = \sigma(\alpha_{sig}\bar{r}_i(k) - 0.5) \\ \omega_i^{sig}(k) = (1-\beta_{sig})\omega_i^{sig}(k-1) + \beta_{sig}\overset{sig}{\omega_i}(k) \end{cases} \tag{17}$$

where α_{sig}=4.0, β_{sig}=0.85, r_0=1.0. Here, r_0 indicates that the transition center is located near a normalized characteristic unit, and a larger α_{sig} is used to produce a more pronounced yet still smooth nonlinear transition, whereas β_{sig}=0.85 preserves continuity in the weight update while allowing a more responsive dynamic variation. The three fixed-weight schemes in the figure are the distortion-prioritized scheme [0.30,0.30,0.30,0.10], the balanced scheme [0.25, 0.25, 0.25, 0.25], and the MTF-prioritized scheme [0.20, 0.20, 0.20, 0.40].

Fig. 9 shows the evolution curves of the composite evaluation value for our method, the two adaptive weighting baselines, and the three fixed-weight schemes. The results show that our method outperforms the fixed-weight schemes, the linear adaptive weighting method, and the Sigmoid adaptive weighting method in terms of convergence speed and final composite evaluation value. Among the linear and Sigmoid schemes, the linear method is faster in the coarse-alignment stage, whereas the Sigmoid method exhibits a clearer MTF response in the later stage, but its overall performance is still inferior to that of our method. These results indicate that the advantage of our method does not simply come from dynamic weight variation itself, but from the fact that its coordination process simultaneously considers the stage-dependent dominant-image-quality regularities identified in Section 2.3 and the distortion-mode-to-DOF association established in Section 2.4, thereby enabling effective concentration on the dominant error source at different alignment stages.

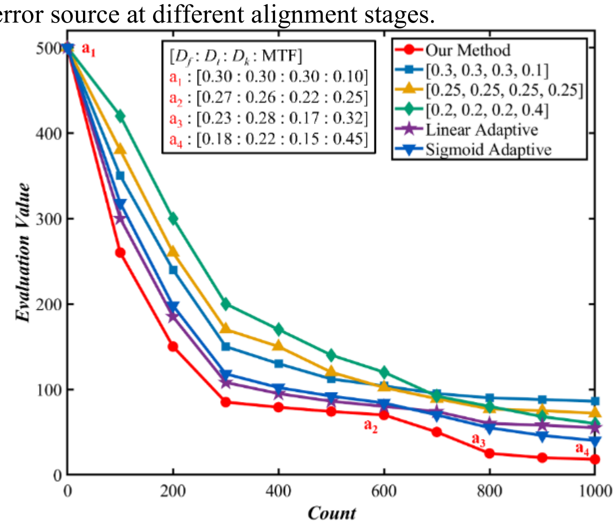

Fig. 9. Comparison of alignment strategies and convergence behaviors.

This subsection further designs two ablation experiments to disentangle the roles of global dynamic weighting and local decoupling in the hierarchical weighting mechanism, and the corresponding results are shown in Fig. 10. In the first ablation, a configuration with globally fixed and locally adaptive weights is adopted. While this setting achieves faster convergence in the early and middle stages than the fully fixed-weight baseline, it quickly enters a plateau in the fine-alignment stage, resulting in only limited final improvement. In the second ablation, globally adaptive and locally fixed weights are employed. This configuration attains a significantly lower final residual than the fully fixed-weight scheme, but its convergence speed in the coarse stage is relatively slow. Overall, the local mechanism mainly accelerates convergence, whereas the global mechanism determines the attainable final accuracy. Their coordinated operation enables rapid convergence and superior terminal performance throughout the alignment process.

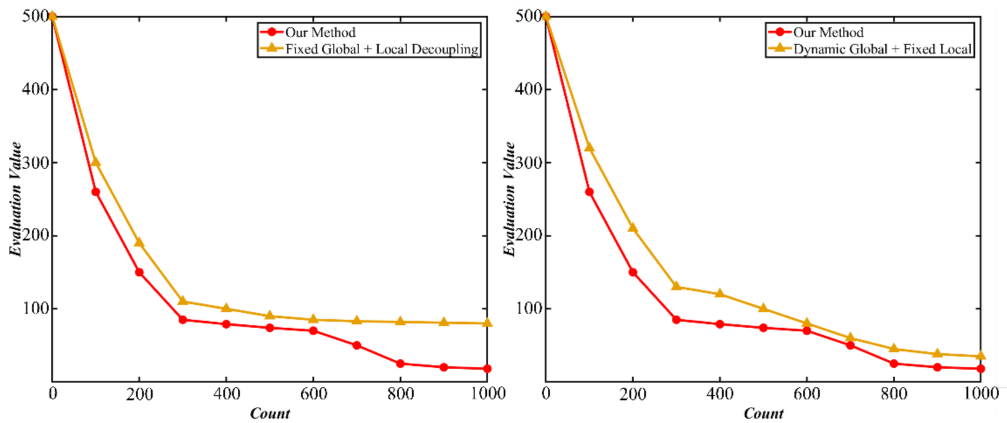

Fig. 10. Ablation Study on Global–Local Weighting Strategies. (a) Comparison between our method and the scheme with fixed global weights and locally adaptive weights. (b) Comparison between our method and the scheme with globally adaptive weights and fixed local weights.

4.3 Robustness Evaluation

To assess the robustness of the proposed automatic alignment scheme with respect to different initial poses, the closed loop alignment procedure was repeated for ten distinct initial poses under the same alignment workflow and a unified evaluation criterion, and the final convergence level and its dispersion were statistically evaluated, as shown in Fig. 11(a). For the proposed method, the composite evaluation value lies in the range 15.08–27.14, with a mean of 20.60 and a standard deviation of 4.03. The relatively concentrated distribution of this composite metric across different initial poses indicates that the method is insensitive to perturbations in the initial state and can consistently achieve stable convergence. In addition, the duration of a single alignment run is kept within 10 min, demonstrating good cycle consistency and repeatability.

Furthermore, the key image-quality metrics are statistically summarized. Fig. 11(b) presents the box plots of the three geometric distortion indices (D_t, D_f, and D_k) and the MTF at 93 lp/mm. The mean values of D_t, D_f, and D_k are 4.86, 14.25 and 18.94, respectively, and the mean MTF is 72.29%. For ease of comparison with the distortion values within the same plot, the MTF is expressed on a percentage scale (multiplied by 100). Overall, the distributions of all metrics are concentrated and show small dispersion, indicating that the algorithm can achieve consistent geometric fidelity and sharpness under different initial poses. Further considering that automatic alignment and manual alignment correspond to pairwise comparisons under the same initial pose, a paired statistical analysis was performed on the overall evaluation value. The results show that, compared with manual alignment, automatic alignment reduced the overall evaluation value by an average of 15.66, with a 95% confidence interval of [9.20, 22.12], and the difference was statistically significant (paired t-test, $t(9) = 5.49$, $p < 0.001$). At the same time, the corresponding effect size was $d_z = 1.74$, indicating that this improvement is not only statistically significant, but also has a strong practical effect. Combined with the small inter-group variation in Fig. 11(a), these results indicate that the proposed method not only achieves better average performance, but also shows good repeatability and robustness.

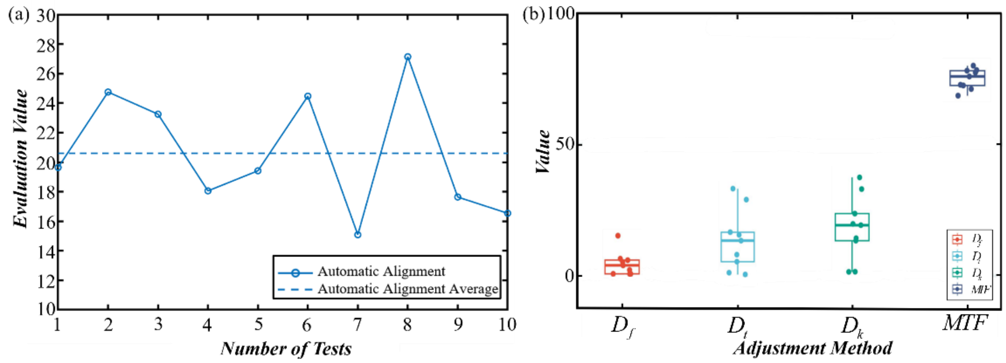

Fig. 11. (a) System Robustness Evaluation. (b) Statistical Comparison of Manual and Automatic Assembly and Adjustment Metrics.

We select three results obtained with the proposed progressive alignment method under different initial poses (Fig. 12(a_1)–(c_1)) and compare them locally with a representative result of manual alignment (Fig. 12(d_1)). From the enlarged details of the two regions of interest (ROIs) located at the upper-right and lower-left corners, it can be seen that, with the proposed method, the horizontal and vertical line pairs at 93 lp/mm are clearly resolved, with sharp and uniform line edges and no sticking or aliasing in the double outer frame lines. In the same regions, the tick marks and fine line segments exhibit well-balanced contrast without noticeable trailing or line broadening. By contrast, in the manually aligned case, pronounced trailing and anisotropic blur appear in the upper-right corner, and the 93 lp/mm line pairs merge. In the lower-left corner, tilt and keystone distortion as well as edge thickening are observed, and the double outer frame lines locally approach each other or even stick together. Taken together, these comparisons show that the proposed method delivers markedly better performance than manual alignment in both high spatial-frequency resolvability and geometric fidelity, and that it provides consistent and stable image quality under different initial poses.

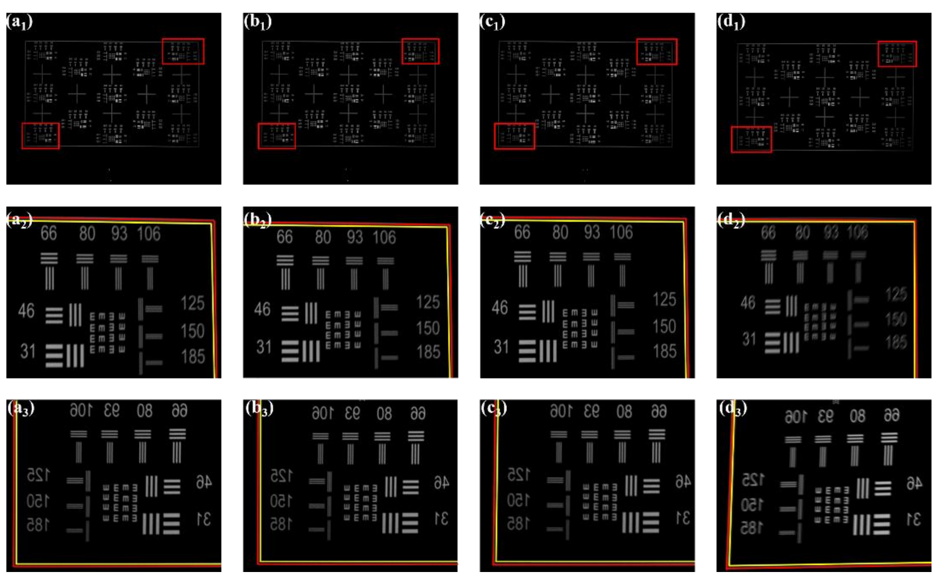

Fig. 12. Comparison of projection imaging between manual and automatic alignment. (a_1–c_1) shows automatic alignment results for three sets of initial poses, while (d_1) shows manual alignment results; red boxes indicate sampling regions. (a_2–d_2) shows enlarged views of the

corresponding upper-right region. (a_3–d_3) shows enlarged views of the corresponding lower-left region.

5. Conclusion

This work focuses on the automatic alignment of a freeform mirror in a catadioptric ultra-short-throw projection system. It investigates the coupling relationships among six-degree-of-freedom pose errors, geometric distortion, and MTF, and on this basis proposes an adaptive weighting alignment method based on the analysis of optical response laws. Simulation results show that, under pose-error perturbations, the dominant roles of geometric distortion and MTF in system performance undergo a stage-dependent shift with the error scale. At the same time, different geometric distortion modes exhibit stable dominant correlations with several subsets of degrees of freedom. Based on the above findings, a stage-wise global weight scheduling strategy and a local grouped optimization mechanism were further established, thereby realizing closed-loop automatic alignment optimization for freeform USTP systems. Experimental results show that the proposed method achieves stable convergence under multiple groups of different initial conditions, with a final average MTF above 70%, an alignment time of about 10 min per run, and significantly reduced result dispersion. Ablation experiments further show that the global weight scheduling strategy and the local grouping mechanism play complementary roles in convergence performance and final alignment quality. Their combination helps improve the efficiency, stability, and repeatability of the alignment process. These results indicate that the weight evolution and optimization organization constructed from the analysis of the pose-image-quality relationship can effectively support the automatic alignment of freeform USTP systems within the current system configuration and alignment range.

Overall, the significance of this work lies not only in achieving efficient automatic alignment for freeform USTP systems, but also in showing that, for strongly coupled optical alignment problems, combining pose-image-quality response analysis of a specific system with the use of physically meaningful dominant laws for stage division, metric scheduling, and optimization organization is an engineeringly feasible technical route. At the same time, the construction of degree-of-freedom grouping and local optimization in this work is still mainly based on the pose-perturbation response characteristics of the current system, and its applicability under more general coupling conditions still requires further investigation. Future work will combine multivariable coupled perturbation analysis, richer initial misalignment scenarios, and validation on other complex optical systems to further improve the theoretical support and application scope of the proposed method.

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